

WORK SHEET - 01(B)

ERRORS IN MEASUREMENTS & INSTRUMENTS

HINTS & SOLUTIONS

Q.4 (B)

$$I = K \tan \theta \therefore \frac{dI}{I} = \frac{\sec^2 \theta}{\tan \theta} d\theta = \frac{2d\theta}{\sin 2\theta}$$

Q.15 (C)

$$x_0 = y_0 z_0$$

$$x = 1.1 y_0 \cdot 0.9 z_0$$

$$\frac{x - x_0}{x_0} = (1.1 \times 0.9 - 1) \times 100 = -1\%$$

Q.21 (ABCD)

$$\frac{\lambda}{4} = l_1 + d \quad [d \rightarrow \text{end correction}]$$

$$\frac{3\lambda}{4} = l_2 + d$$

$$\therefore \frac{\lambda}{2} = l_2 - l_1 \Rightarrow \lambda = 2(l_2 - l_1)$$

$$\therefore \lambda \text{ may be equal to } [2(39.3 - 13)] \text{ to } [2(39.1 - 13.2)]$$

$$\lambda \text{ may range from } 51.8 \text{ cm to } 52.6 \text{ cm}$$

Q.22 $\frac{4}{\pi} \sqrt{3} \times 10^{-2}\% \left(\frac{0.12}{\sqrt{3}\pi} \% \right)$

Q.23 50 cm

Q.24 $[(10.1 \pm 0.2) \text{ ms}^{-2}]$

Q.25 [5]

$$\frac{\Delta y}{y} = 2 \times \frac{0.01}{1.25} \times 100 + \frac{.001}{0.125} \times 100 + \frac{0.01}{50} \times 100 = 2.42$$